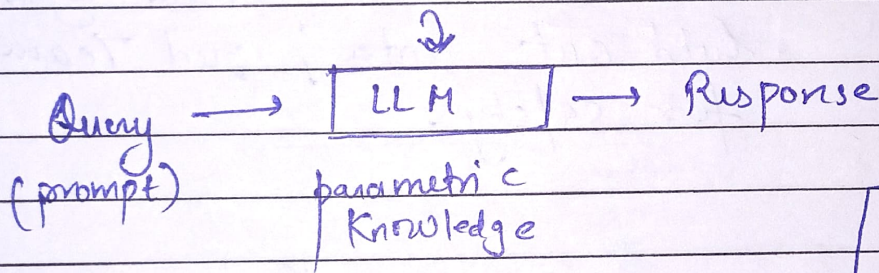
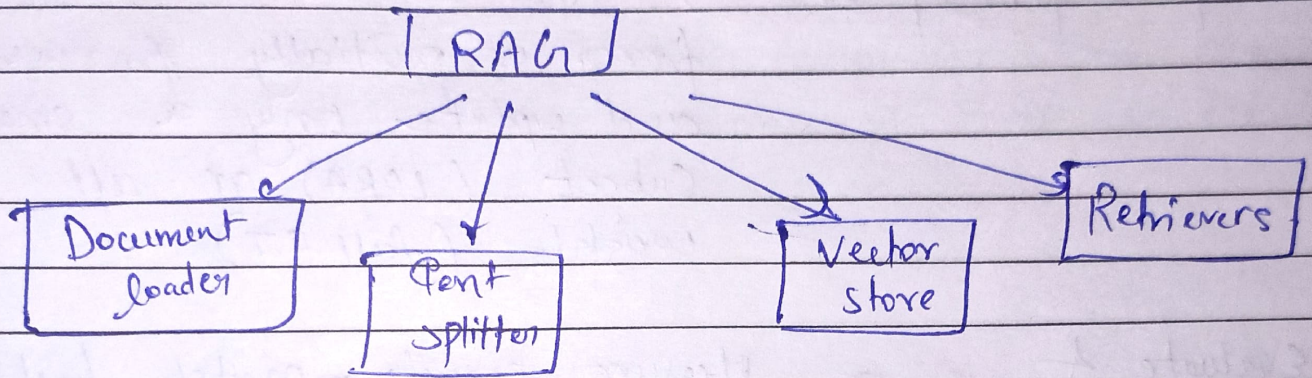
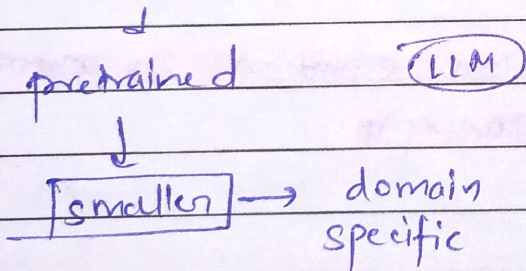


# RAG ( Retrieval Augmented Generation ) :

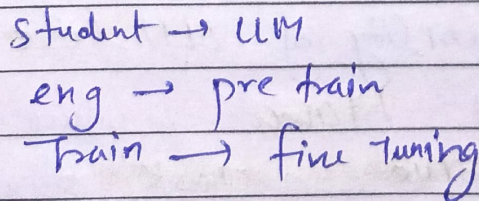


- 1) private data
- 2) recent data
- 3) Hallucination

## fine Tuning



or



supervised  
Tuning

↓  
labelled dataset

prompt → desired output

Unsupervised (continued pretrained)

↓  
Transcript

→ Model

~~RLL~~ RLL

(Reinforcement Learning Human feedback)

# # Fine Tuning process

Date .....

1. collect data → labelled data.
2. choose a method → full parameters, LORA/QLORA, or parameter efficient adapters.
3. Train for a few epochs → you keep the base weight frozen or partially frozen and update only a small subset (LORA) or all weights (full FT)
4. Evaluate & Safety-Test → Measure exact-match factuality and hallucination rate against hold out data; red Team for safety.

- 
- 1) LLM → Train
  - 2) technical expertise
  - 3) computationally expensive
- problem in fine tuning.
- (RAG is cheaper solution as compared to fine tuning)

## ★ Solution In Context Learning :-

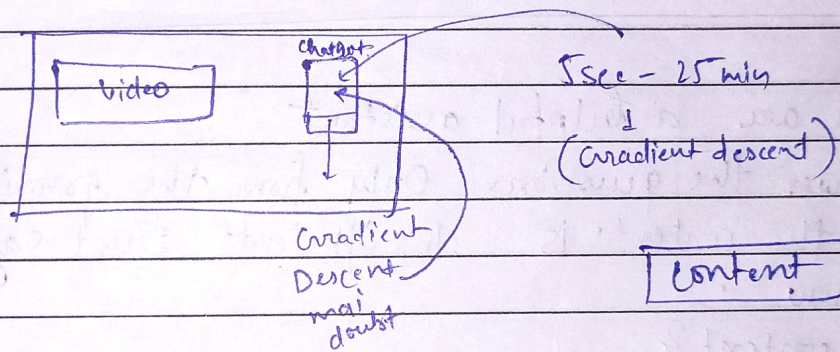
↳ is a core capability of LLM like GPT 3/4, Claude and Llama, where the model learn to solve a task purely by seeing examples in the prompts without updating its weights.

LLM → emergent property hai

↳ In content learning.

→ An emergent property is a behaviour or ability that suddenly appears in a system when it reaches a certain scale or complexity - even though it was not explicitly programmed or expected from the individual components.

for ex:



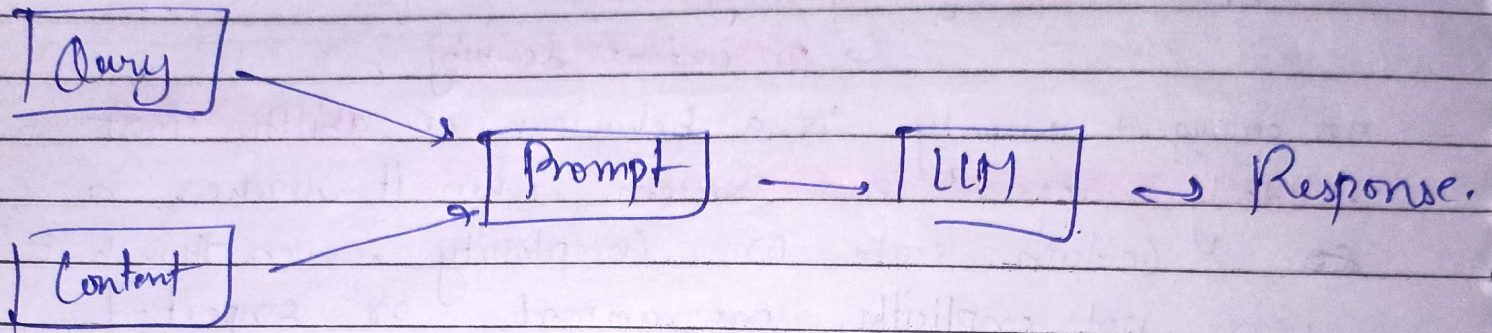
∴ [means pura ka pura content inject kr rhe hai prompt ke andar]

→ Instead of just example Task, retrieve background Info, facts, documents, product manuals etc.

→ Injects that into the prompt to augment the model's Knowledge.

RAG is a way to make a language model (like GPT) smarter by giving it extra information at the time you ask your question.

↳ Definition RAG.



Prompt :

" " "

You are a helpful assistant  
 Answer the questions Only from the provided Content  
 If the content is Insufficient Just say you don't  
 know

{ content }

Question : { Question } " " "

#

## Understanding RAG

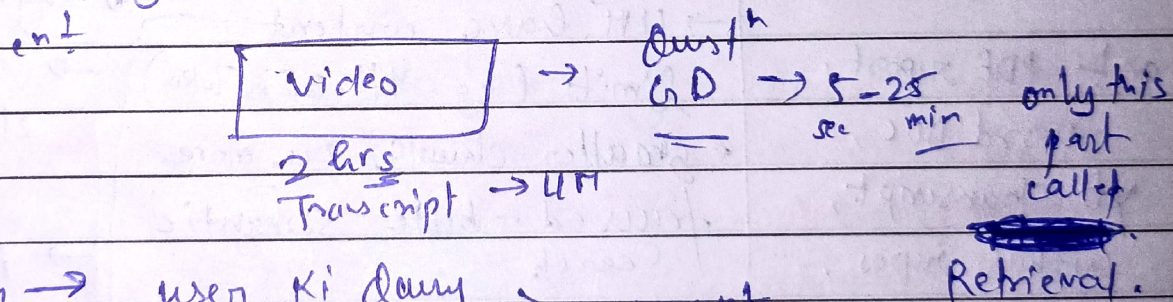
RAG → Information Retrieval + text generation.

- Indexing
- Retrieval
- Augmentation
- Generation

→ 4 steps in RAG

external knowledge base.

- Indexing → external Knowledge<sup>base</sup> provide to LLM (as content)
- Retrieval → Query ko samjhate hai aur external Knowledge base mai Jaake un Chunk ko dhoodhte Jo Query ko answer Krne mai help Krnge.



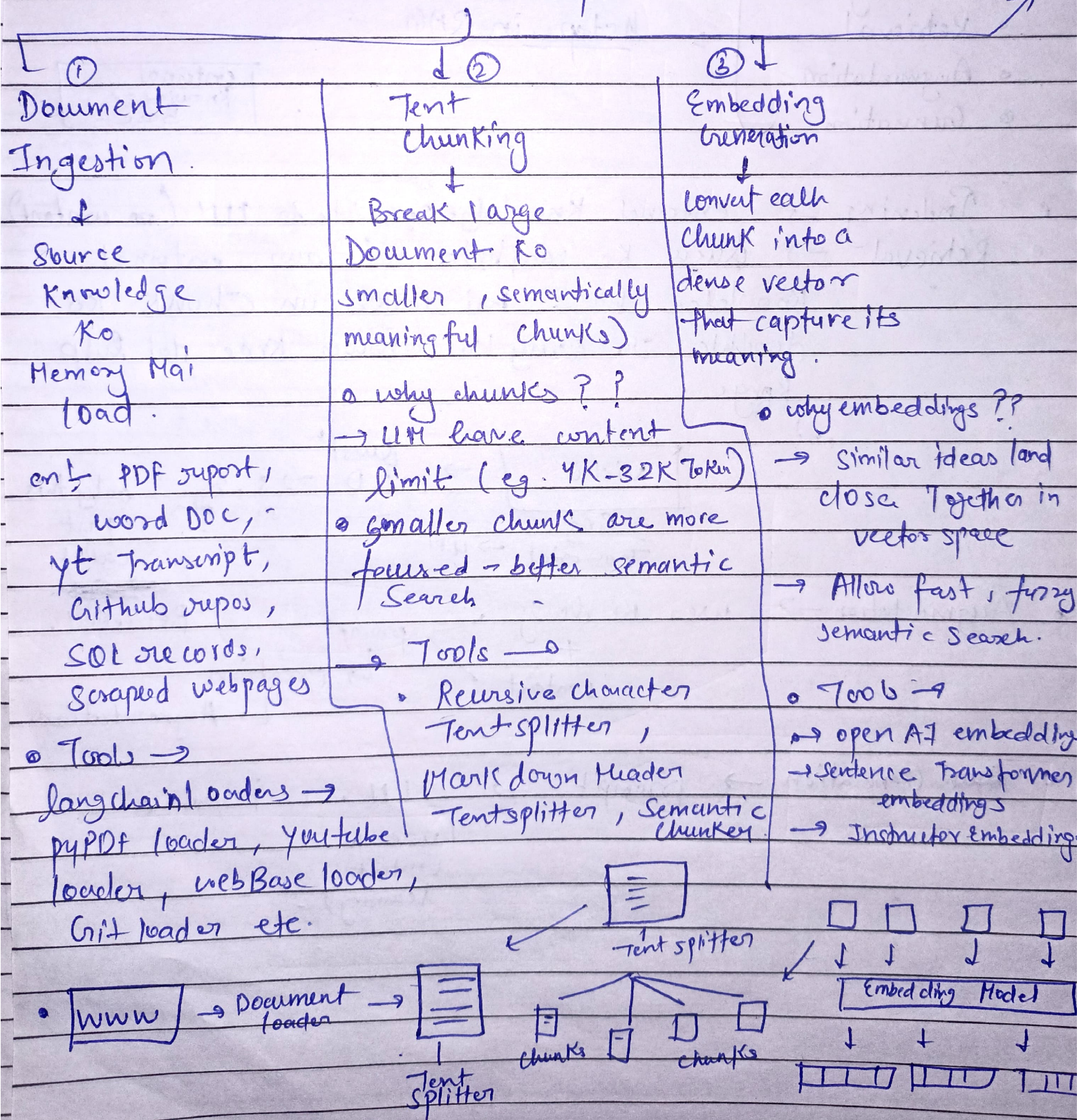
- Augmentation → user ki Query + content → prompt  
 ↳ creation  
 ↳ Augmentation.

- ~~Gen~~ Generation → prompt → LLM → Response.

(use In content learning) -

① • Indexing → Indexing is the process of preparing your knowledge base so that it can be efficiently searched at query time.

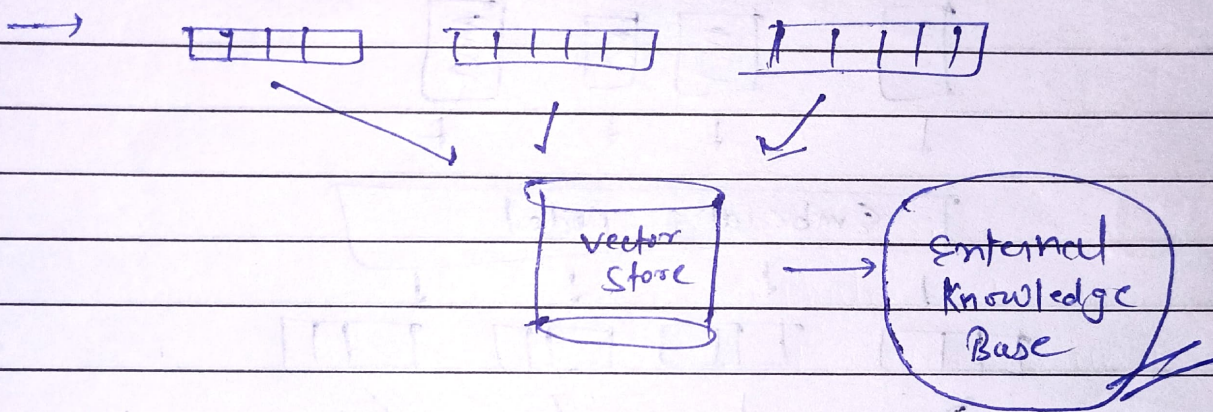
→ 4 sub steps.



④ Storage in vector store → store the vector along with the original chunk Text + metadata in a vector database

→ Vector DB options:

- local → FAISS, chroma
- cloud → pinecone, Weaviate, Milvus, Qdrant



② Retrieval → is the real time process of finding the most relevant piece of information from a pre-built Index (created during Indexing) based on the User's Question.

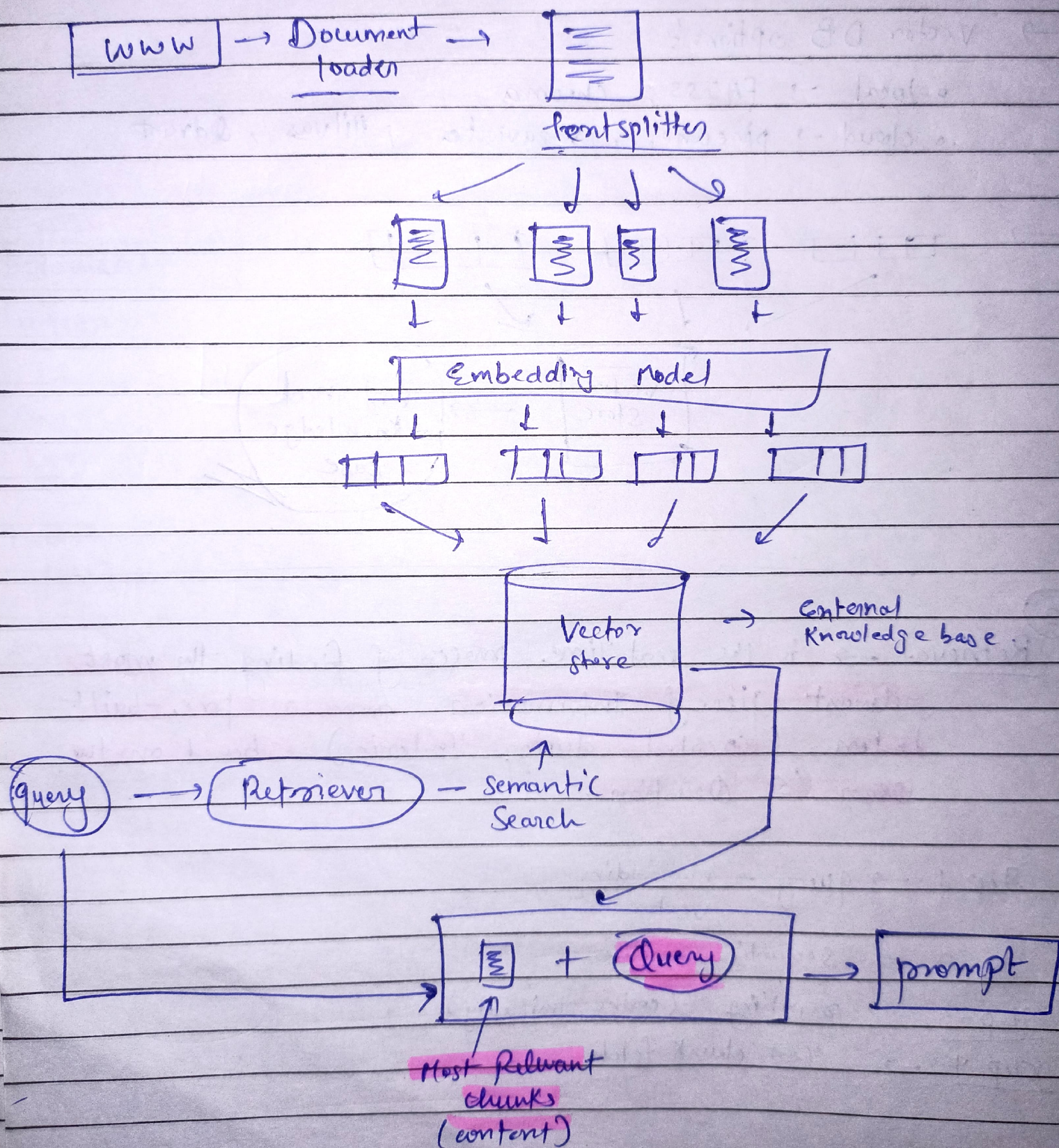
Step 1 → query → embedding Vector

Step-2 → Semantic Search.

Step-3 → ranking (cosine similarity)

Step-4 → Text chunk fetch.

### ③ ~~Generation~~ Augmentation :-



Date .....

- ⑥ Generation  $\hat{=}$  final steps where a LLM uses the user's query and the retrieval & Augmented content to ~~the~~ generate a Response.

prompt  $\rightarrow$  [ LLM ]  $\rightarrow$  Response

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